region) towards the poles (energy deficit regions) making up for heat loss at higher latitudes.

And most of the heat transfer takes place across the mid-latitudes (30° to 50°) [more while studding jet streams and cyclones], and hence much of the stormy weather is associated with this region.

39. **Statement 1: CORRECT**—With increase in elevation, the atmospheric pressure falls. Fall in pressure implies that the temperature also falls [Pressure is directly proportional to temperature and vice versa] Statement 2: CORRECT—With increase in elevation, the concentration of greenhouse gases decrease (Water vapor and carbon dioxide fall sharply with elevation). Hence the heat absorption capacity of atmosphere will also decrease.

    This sort of fall in temperature with elevation is called Temperature Lapse and the rate at which it happens is called Temperature Lapse Rate or simply Lapse rate.

40. Equatorial Low Pressure Belt: This belt happens to be the zone of convergence of trade winds from two hemispheres from sub-tropical high pressure belts. This belt is also called the Doldrums, because of the extremely calm air movements.

    **Sub-tropical High Pressure Belt**: The corresponding latitudes of sub-tropical high pressure belt are called horse latitudes.

    In early days, the sailing vessels with cargo of horses found it difficult to sail under calm conditions of this high pressure belt.

    They used to throw horses into the sea when fodder ran out. Hence the name horse latitudes.

    **Sub-Polar Low Pressure Belt** Located between 45°N and S latitudes and the Arctic and the Antarctic circles (66.5° N and S latitudes).

    On long-term mean climatic maps, the sub polar low- pressure belts of the northern hemisphere are grouped into two centers of atmospheric activity: the Iceland low and the Aleutian depression (Aleutian low).

41. There are two main causes, thermal and dynamic, for the pressure differences resulting in high and low pressure systems.

    **Thermal Factors**: When air is heated, it expands and, hence, its density decreases. This naturally leads to low pressure. On the